

WEEK 1 – UNDERSTANDING THE BUSINESS PROBLEM

Requirement Analysis Document: Online Shopping Platform

1. Introduction

An online shopping platform is an e-commerce system that allows customers to purchase products through the internet. Platforms such as Amazon and Flipkart provide services including customer registration, product browsing, shopping cart management, online payment, order processing, delivery, and shipment tracking.

The proposed system will provide a simple and user-friendly platform where customers can search for products, place orders, make payments, and track their deliveries.

2. Business Problem

Traditional shopping requires customers to physically visit stores, compare products, and make purchases. An online shopping system solves these problems by allowing customers to shop from anywhere and at any time.

1. Easy customer registration and login.
2. Convenient product search and browsing.
3. Shopping cart and wishlist facilities.
4. Secure order placement and payment.
5. Order confirmation and shipment tracking.
6. Admin facilities for managing products, customers, and orders.

3. Major Activities in an E-Commerce System

1. Customer Registration – Customers create an account using their basic details.
2. Customer Login – Registered customers securely log in to the system.
3. Product Browsing – Customers view available products and their details.
4. Product Search – Customers search for products using keywords, categories, or filters.
5. Shopping Cart – Customers add, remove, or update products before purchasing.
6. Order Placement – Customers provide delivery details and confirm their order.
7. Payment Processing – Customers select a payment method and complete the transaction.
8. Order Processing – The system records and processes the customer's order.
9. Shipment – The ordered product is packed and dispatched for delivery.
10. Shipment Tracking – Customers track the current status of their orders.
11. Order Delivery – The product is delivered to the customer's address.
12. Feedback and Reviews – Customers can provide ratings and reviews for purchased products.

4. Functional Requirements

1. The system shall allow customers to register and create an account.
2. The system shall allow registered customers to log in and log out.
3. The system shall display products with name, price, description, images, and availability.
4. The system shall provide product search and filtering options.
5. The system shall allow customers to add products to a shopping cart.
6. The system shall allow customers to modify or remove items from the cart.
7. The system shall allow customers to enter and manage delivery addresses.
8. The system shall allow customers to place orders.
9. The system shall provide available payment options.

10. The system shall generate an order confirmation after successful order placement.
11. The system shall allow customers to view their order history.
12. The system shall provide shipment tracking information.
13. The system shall allow customers to cancel eligible orders.
14. The system shall allow customers to submit product ratings and reviews.
15. The admin shall be able to add, update, and remove products.
16. The admin shall be able to manage customer orders and update order status.

5. Non-Functional Requirements

1. Usability – The system should have a simple and user-friendly interface.
2. Performance – Product pages and searches should load quickly.
3. Security – Customer and payment-related information must be protected.
4. Reliability – The system should work correctly without frequent failures.
5. Scalability – The system should support increasing numbers of customers and products.
6. Availability – The platform should be accessible whenever customers need to shop.
7. Maintainability – The system should be easy to update and maintain.

6. Main Users of the System

1. Customer – Searches products, manages the cart, places orders, makes payments, and tracks shipments.
2. Admin – Manages products, customers, orders, payments, and system activities.
3. Delivery Personnel – Handles product delivery and updates shipment status.

7. Basic System Workflow

Customer Registration/Login → Browse/Search Products → Select Product → Add to Cart → Checkout → Enter Delivery Details → Payment → Order Confirmation → Order Processing → Shipment → Tracking → Delivery → Review

8. Expected Outcome

The proposed online shopping system will provide customers with a convenient and efficient way to purchase products online. It will reduce the need for physical shopping and provide important features such as product search, online ordering, payment processing, and shipment tracking.

9. Conclusion

The requirement analysis identifies the major functionalities and users of an online shopping platform. These requirements will serve as the foundation for designing and developing the proposed e-commerce system in the following stages of the project.